PAP521S – PROGRAMMING IN PRACTICE

Project A: Municipal Financial Management System (MFMS)

Short Technical Report – Final

Group Members

No.	Group Member	Student Number
1	Magdalena Jakob	226037967
2	Paul Esau	220087318
3	Hendrina Shifeleni	226164519
4	Elago Nilingene	226016153
5	Katijume Many	226048373
6	Shiimi Catherine	226094901
7	Ryan Kamolakakame	226051625

GitHub Repository: https://github.com/OgaleCode/MFMS

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1. Introduction

This report presents the foundation version of the Municipal Financial Management System (MFMS), developed in ANSI C (C99) using GCC in Visual Studio Code. The purpose of the system is to manage municipal records as part of Project A of PAP521S – Programming in Practice.

2. Problem Description

Manual record books can lead to data loss and incorrect calculations. The municipality therefore needs a C program that follows the flow of input, validation, storage, search, calculation and output, and that checks the data entered by the user.

3. System Objectives

- Build a menu-driven C application that applies variables, operators, if/switch decisions, loops, arrays, strings and functions.
- Manage employees, budgets, suppliers and assets.
- Provide reports.
- Work as a team of seven using GitHub.

4. System Features

Module	Description	Developed by
Main Menu	Six options, handled with a switch statement.	
Employee Management	Add, display and search employees (search uses strcmp). Salary = basic salary + housing allowance + transport allowance.	Magdalena
Budget Management	Remaining budget = allocated budget – expenditure. Status is shown as WITHIN or EXCEEDED.	Hendrina
Supplier Management	Stores supplier ID, name, email, phone and town.	Elago
Asset Management	Stores asset ID, name, type, value, department and condition.	Marty
Reports	Average, highest and lowest salary, and budget totals.	Catherine and Ryan
Integration	Program organised into modular functions.	Paul

5. Program Design

Files: main.c, employees.c/.h, budget.c/.h, suppliers.c/.h, assets.c/.h and reports.c/.h.

Data records: Data records are stored in arrays of structures, for example an array of struct Employee with space for 100 records.

String functions: strlen(), strcmp() and strcpy() are used where the program needs them.

6. Challenges Encountered

- Using scanf with input that contains spaces caused a segmentation fault.
- main() became too long.
- Git merge conflicts occurred when members worked on the same files.
- Using == to compare strings instead of strcmp().
- Negative values needed to be rejected by validation.
- Calculating an average on an empty array.

7. Solutions Implemented

- The code was split into separate modules and functions so that main() only displays the menu and calls the module functions.
- Strings are compared with strcmp() instead of ==.
- Validation checks reject negative values.
- The average is only calculated when the array contains records.
- Work was committed regularly to GitHub, and merge conflicts were resolved before integration.

8. Conclusion

The MFMS foundation version provides a menu-driven C program for managing employees, budgets, suppliers and assets, with validation, search and reports. The project applied the concepts covered in the first part of the course and gave the group experience working as a team on GitHub. The modular structure allows the system to be extended in Project B.